Key: Four Million People Who Are Not in the Denominator

84-355 Democracy's Data | Felony disenfranchisement | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure, as of the rebuild. The brief (`disenfranchisement-brief.Rmd`) is now a chapter rather than a parallel copy of the lab: prose organised by argument, under idea headings, with three numbered figures and no numbered steps. The lab (`disenfranchisement.Rmd`) still runs Steps 1-10 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab**; the chapter is cited by figure number or section title.

Teach this near the turnout labs. It is the missing piece of every turnout rate in the course — the CPS lab's 64.4%-or-58.5% gap is partly *this*. If you run it right after, students can go back and recompute.

It is also the shortest lab in the set. Two tables and a bar chart. The time goes on Steps 8-9 and the discussion.

Every number below was run against the data files.

The session in one sentence

4,049,978 Americans cannot vote because of a felony conviction, 40% of them have finished their sentences, and whether they are in your turnout denominator is a choice somebody made without telling you.

The findings

```
c(total = n$total, post_sentence = n$post_sentence,
  pct_post = round(100 * n$post_sentence / n$total),
  national_rate = n$pct,
  states = nrow(s))
```

```
#>         total post_sentence      pct_post national_rate        states
#>     4049978.0     1616437.0          40.0           1.7          50.0
```

```r
head(s[, c("state", "total", "pct")], 4)
```

```
#>       state  total  pct
#> 1 Tennessee 399684 7.68
#> 2   Florida 961757 6.13
#> 3   Alabama 227437 5.95
#> 4  Kentucky 158381 4.66
```

- **4,049,978 disenfranchised, 1.70% of the voting-eligible population.**
- **1,616,437 — 40% — are post-sentence.** The largest single category.
- **Tennessee 7.68%**, Florida 6.13%, Alabama 5.95% — the top rate is 5.0× the average among the states that bar anyone at all.
- **Maine and Vermont: zero.** They are the only 2 states that bar nobody at all — people vote from prison.

The three things to get said

1. The biggest group has finished being punished. Students arrive assuming this is about people in prison. Prison is 1,034,995 of 4,049,978 — 26%. **Post-sentence is the largest category**, and it is entirely a policy choice made after the court is done. Put those two numbers next to each other on the board before anything else.

2. The variation is law, not crime. Tennessee's rate is not high because Tennessee has more crime than Vermont. **It is high because Tennessee keeps the penalty running and Vermont never imposes it.** Ask what it means that the same conviction produces permanent disenfranchisement in one state and none in another.

3. Then close the loop to the denominator. This is why the lab is in a data course rather than a criminal-justice one. **Every turnout rate the class has computed has silently included or excluded these people**, and the CPS lab already showed a 5.9-point swing from denominator choice. Have someone state which denominator they would use and defend it.

The line for the board: a turnout rate is an opinion about who counts as a voter.

The methods point worth two minutes

The source is a PDF. No spreadsheet exists; these numbers live in Table 2 on page 18 of a report. The build extracts them and then **checks the parse** — every state's five categories must sum to its total.

That check earned its keep. Texas reports 139,631 and 100,600, wide enough to overflow their column, and a naive fixed-width read splits the numbers in half. The sum test caught it; without the test the lab would have shipped with Texas quietly wrong or missing.

Worth saying out loud: **the check is what makes an extracted table usable.** Students will extract data from PDFs for the rest of their lives.

Handling the room

Charged, and in a different way from the discrimination unit — this one has a live policy argument attached with people on both sides in most classrooms.

- **The lab does not take a side and neither should the session.** The finding is the 40% and the state variation; what to do about it is the discussion.
- **Head off the electoral-consequences argument early** — see Option C. Students will want to say a state's disenfranchised population exceeds some margin, and that comparison needs care.

Walking the steps

Steps	What they do	Min	Watch for
1-3	A PDF source, the sum check, and one state's row	10	**43 states sum exactly, 7 are off by one.** That is rounding, not a parse failure — and the distinction is the lesson.
4-5	4.05m people; then the post-sentence column	14	**The pivot.** 71.5% not incarcerated; 39.9% finished their sentence; 10 states hold all of it.
6-7	The range across states, then the top of the table as a bar chart	9	Tennessee 7.68% against Maine and Vermont at zero. The chapter's **Figure 3** is the cartogram of the same thing; the lab plots bars.
8-9	The denominator is itself an estimate; the summary	7	Adult citizens, not current electorate. Say which.
10	What you can say	5	No demographics in this file — the racial disparity is outside it.

Step 5 is the pivot. Post-sentence is the largest category, larger than prison and jail combined, and it exists in only 10 states. Everything political about this topic is in that one column.

If you are short on time, protect the pivot and the closing step. The numbers are what make the argument concrete; the sequence is the argument.

Option A — Rank by the right thing

Verified output

```
head(s[order(-s$total), c("state", "total", "pct")], 6)
```

```
#>        state  total  pct
#> 2    Florida 961757 6.13
#> 10     Texas 479097 2.50
#> 1  Tennessee 399684 7.68
#> 6   Virginia 264292 4.18
#> 8    Georgia 249699 3.25
#> 3    Alabama 227437 5.95
```

By rate it is Tennessee, Florida, Alabama. By count it is Florida, Texas, Tennessee — and Texas, second by count, is 10th by rate at 2.50%, against 7.68% at the top.

3/4: Produces both rankings and notes Texas jumps.

4/4: Says which serves which argument. **Rate** measures how much a state's electorate is diminished; **count** measures how many people are affected. Someone arguing this is a national problem rather than a Southern one reaches for Texas, California and the total — California is 9th by count and only 35th by rate; someone arguing about the health of a particular democracy reaches for the rate.

4/4, best: Notices neither ranking is neutral and both are honest — and that the choice between them is the same choice as the denominator question in **Steps 8-9**. **The unit you divide by is an argument.**

Option B — Abolish post-sentence disenfranchisement

Verified output

```
c(national_now = n$pct,
  national_after = round(100 * (n$total - n$post_sentence) / n$voting_eligible_pop, 2),
  states_above_1pct_now = sum(s$pct > 1),
  states_above_1pct_after = sum(s$new_pct > 1))
```

```
#>            national_now          national_after   states_above_1pct_now
#>                    1.70                    1.02                   19.00
#> states_above_1pct_after
#>                   18.00
```

```
s$drop <- s$pct - s$new_pct
head(s[order(-s$drop), c("state", "pct", "new_pct", "drop")], 5)
```

```
#>       state  pct new_pct drop
#> 1 Tennessee 7.68    1.91 5.77
#> 3   Alabama 5.95    1.21 4.74
```

```
#> 2   Florida 6.13    1.48 4.65
#> 6  Virginia 4.18    1.50 2.68
#> 5   Arizona 4.20    2.01 2.19
```

The national rate falls from 1.70% to 1.02% — a 40% reduction, by definition. But the state effects are wildly uneven:

- **Tennessee 7.68% → 1.91%**, a fall of 5.77 points
- **Alabama 5.95% → 1.21%**; **Florida 6.13% → 1.48%**
- **Arkansas, Georgia, Texas and Idaho do not move at all**

3/4: Recomputes and reports the national fall.

4/4: Notices that **40 of 50 states report zero post-sentence disenfranchisement**, so the reform would do nothing in most of the country and almost everything in a handful of states. The policy is not national; it is about 10 states.

4/4, best: Sees that this reorders the table. After the change, the highest rates are **Arkansas and Georgia**, which ranked 7th and 8th before — because their disenfranchisement is concentrated in probation, which is unaffected. **A reform aimed at the visible problem leaves a different set of states at the top**, and a student who notices that the ranking is an artefact of which mechanism you target has understood the structure.

Option C — Put it against a margin

No verified output; **grade the care, not the comparison.**

3/4: Finds a race with a margin smaller than the state's disenfranchised population and states the comparison.

4/4: Immediately qualifies it correctly. Disenfranchised people are **not a random sample** — they are disproportionately young, poor, and Black, all groups with turnout below average, and formerly incarcerated people who regain the vote turn out at low rates. **A population of 400,000 does not become 400,000 votes.**

4/4, best: Says what the comparison *is* good for. It establishes that the policy operates at a scale where it *could* matter, which is a claim about magnitude and is defensible. It does not establish that any election would have gone differently, which is a counterfactual this data cannot support. **A student who separates "large enough to matter" from "decided the outcome" has the answer** — and it is the same distinction the RPV lab needed between direction and magnitude.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is connecting the four million to a denominator they have already used in this course.

- **A 2** is reporting the state ranking as a fact about crime.
- **Also a 2**: asserting an election would have flipped. See Option C.
- **Reward anyone who asks about the estimates' uncertainty.** These are modelled figures published without intervals, and noticing that is right.

Anticipated questions

- *"Are these people counted in the census?"* — Yes, and mostly **where they are incarcerated**, not where they lived. That is prison gerrymandering, and it is the same population appearing in a different lab's problem.
- *"Can they get the vote back?"* — Depends entirely on the state: automatic at release, at end of parole, after fees are paid, or only by pardon. Florida's 2018 amendment and the fee litigation that followed is the case students will have heard of.
- *"Why is DC missing?"* — Not in Table 2 of the source. Say so plainly.
- *"Is 1.70% a lot?"* — It is 1 in 59 adults nationally and 1 in 13 in Tennessee. **Give both; the national figure conceals the policy.**
- *"Who produced these numbers?"* — Academic researchers, from correctional counts and state law. Not a government count — **nobody official publishes this**, which is itself worth a beat.

Connections

- **CPS turnout** — the 64.4%/58.5% denominator gap, partly explained here.
- **The census** — where incarcerated people are counted, and the VAP/CVAP/VEP distinction.
- **Section 203 and EAVS** — thresholds and rates that depend on a chosen population.
- **Jury selection and policing** — the same populations, measured at a different stage of the same system.